

Potential Shaping with Changing Coalitions: A Fixed-Population Ledger Repair

Abstract

MARS-RA uses multimodal pairwise comparisons, Bradley–Terry aggregation, and potential-based reward shaping for cooperative multi-agent reinforcement learning with changing active coalitions. Its comparison scores and potentials live in the current coalition space, so consecutive vector potentials need not have the same dimension; it also combines a scalar team reward with a vector shaping term without declaring payoff semantics. We give a narrow repair. Active credits are embedded in a fixed population space, inactive agents take null actions, and every population member retains a reward ledger on every transition. Componentwise telescoping then shifts each declared per-agent payoff by a policy-independent constant, preserving best responses and Nash equilibria for arbitrary comparison outputs. If shaping rewards are paid only while an agent is active, the payments dropped at the agent’s entry and exit transitions are left behind as boundary terms that can depend on policy; we work this out for the two natural payment conventions. Nash invariance under static and time-varying potentials is prior work, and so is the observation that a nonzero potential at an action-dependent terminal state leaves a policy-dependent term; the contribution here is the typed ledger formulation for changing coalitions and the resulting entry and exit diagnosis. We also separate Bradley–Terry sampling error from semantic bias and explain why neither comparison accuracy nor a local signal bound yields a finite-training guarantee without a learner-stability premise.

1 Introduction

Learning-augmented algorithms require a declared prediction interface, an error measure, and a theorem connecting prediction quality to downstream performance. They distinguish consistency and error-sensitive guarantees from prediction-independent robustness, and warn that semantic predictions do not automatically compose into system guarantees [5]. MARS-RA instead uses multimodal pairwise comparisons to fit Bradley–Terry contribution scores and converts their softmax into a potential-based shaping signal for cooperative multi-agent reinforcement learning [4].

The tempting question is how comparison error degrades the limiting return. That question is misplaced under exact potential shaping, by which we mean throughout the fixed-population all-transition ledger of Section 3 with a zero terminal potential and a fixed initial augmented state: there the discounted shaping terms telescope, so their total is a common initial offset. The comparisons may affect optimisation, exploration, or sample efficiency, but comparison accuracy does not by itself yield a finite-training return bound; a stability, regret, or convergence contract for the learner would have to be added, and neither source paper supplies one.

Our result is narrower. MARS-RA defines a fixed population but places each score in the current active-coalition space, then subtracts consecutive vector potentials. This expression is not typed when coalition sizes change. A fixed-population embedding is readily available, but inactive coordinates, transition ownership, and per-agent payoffs must be declared. We provide that construction and show which payments an active-only implementation drops and what they leave behind.

The main steps are not new. Lu, Schwartz, and Givigi prove that potential-based shaping preserves Nash equilibria in general-sum stochastic games [3]. Devlin and Kudenko, in “Dynamic Potential-Based Reward Shaping” (AAMAS 2012), show that the same holds for potentials that change over time [1]; MARS-RA appeals to that result for its own shaping [4]. Grześ, in “Reward Shaping in Episodic Reinforcement Learning” (AAMAS 2017), showed that a nonzero potential at an action-dependent terminal state leaves a term that can alter policies and Nash equilibria [2]. Our contribution is the fixed-population reward-ledger formulation for changing coalitions, together with the observation that omitting inactive-agent ledger entries turns each agent’s entry and exit into boundaries of the shaping sum, where the dropped boundary payments reappear as boundary terms that can depend on policy. We do not claim a new general reward-transformation theorem.

2 What the two papers establish

The survey of Zhao et al. provides a vocabulary and discipline for prediction interfaces, robustness, error propagation, and prediction cost; it does not analyse MARS-RA or prove a MARL learner-stability theorem [5]. Wang et al. define MARS-RA over a fixed population with a changing active coalition. They obtain pairwise multimodal comparisons, aggregate them through a Bradley–Terry estimator, apply softmax, and use the resulting vector in potential-based reward shaping [4].

Their estimator statement (Proposition 1) concerns concentration of the estimate $\hat{c}_t$ around the comparator’s latent Bradley–Terry preference c_t^* . Bradley–Terry scores are identified only up to translation, so c_t^* is a translation class. The contribution target is a different object: v_t^* denotes the Shapley-value vector of the agents’ contributions, which MARS-RA offers as an interpretive reference. Its Proposition 2 assumes $c_t^* = v_t^*$ as an explicit hypothesis and concludes that

$\hat{c}_t$ is a consistent estimator of v_t^* up to a translation constant; MARS-RA frames the proposition as an interpretive lens rather than as recovery of Shapley values in practice. Under that reading, repeated queries can reduce sampling error about c_t^* while any systematic semantic gap between c_t^* and v_t^* remains. Our own reading of Proposition 1 and its appendix proof is that the stated rate is not ready for composition. The proposition normalises the error by $1/\sqrt{n}$ and states a rate of order $\sqrt{n \log n/K}$; the appendix restatement drops the normalisation; and the final assembly of the proof obtains $\sqrt{\log n/K}$. The displayed estimator is an unregularised maximum-likelihood estimate, which may diverge under separation, while the proof sketch invokes the analysis of regularised maximum likelihood; connectedness of the comparison graph is invoked for restricted strong convexity without a uniform curvature bound; and Step 2 of the proof replaces the gradient by a first-order Taylor expansion without an error term. We therefore use no numerical Bradley–Terry rate.

3 Fixed-population formulation

Let the population be $N = \{1, \dots, n\}$ and let $I_t \subseteq N$ be the active coalition at time t . Time runs over $t = 0, 1, \dots, T$, where T is the termination time of the episode, which may depend on the actions taken. Let $\gamma \in [0, 1]$ be the discount factor, $\rho > 0$ the shaping weight, and π the joint policy of the population; $\mathbb{E}_\pi$ denotes expectation over trajectories generated by π . For a vector c indexed by a nonempty set I , $\text{softmax}_I(c)$ is the softmax taken over the coordinates in I . Let $\hat{c}_t$ be any realised score vector indexed by I_t . We permit biased, stochastic, or history-dependent scores. The augmented state is $x_t = (s_t, I_t, h_t)$, where s_t is the environment state and h_t carries whatever information makes the realised potential a function of x_t ; for deterministic scores computed from s_t and I_t , h_t is empty. MARS-RA models its task as an open decentralised partially observable Markov decision process, in which agents act on partial observations [4]. Best responses and Nash equilibria below are stated in the stochastic game on the augmented state, over the policies of the population’s players; because the payoff shift in the theorem holds on every trajectory, it does not depend on the policy class, and the same shift applies to policies over partial observations. Augmentation may be required for a standard stochastic-game equilibrium interpretation; the algebra below does not need it.

For nonterminal t with $I_t \neq \emptyset$, define $\Phi_t \in \mathbb{R}^n$ by

$$\Phi_t^i = \mathbf{1}\{i \in I_t\} \text{softmax}_{I_t}(\hat{c}_t)^i, \quad i \in N, \quad (1)$$

and set $\Phi_T = 0$ at termination. A complete model must separately declare a fallback for an empty coalition; nothing below supplies one. Give inactive agents null actions and maintain a payoff ledger for every $i \in N$ on every transition. For declared per-agent environmental payoff r_t^i , define the repaired team reward

and objective by

$$r_t^{\text{team}} = \sum_{i \in N} r_t^i, \quad J^{\text{team}}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t r_t^{\text{team}} \right]. \quad (2)$$

If these payoffs allocate MARS-RA's original scalar team reward, this definition additionally requires their sum to equal that reward on each transition. Define

$$F_t^i = \gamma \Phi_{t+1}^i - \Phi_t^i, \quad \tilde{r}_t^i = r_t^i + \rho F_t^i. \quad (3)$$

Theorem 1 (fixed-population ledger invariance). *Assume that $I_t \neq \emptyset$ for every nonterminal t and that the termination time T is almost surely finite. Fix the initial augmented state x_0 and set $\Phi_T = 0$. Then, for every realisation of the comparison outputs, the discounted shaping sum of player i under (1)–(3) equals $-\Phi_0^i$, so the construction shifts player i 's discounted payoff by $-\rho \Phi_0^i$. Consequently, it preserves every unilateral payoff difference, best-response correspondence, and Nash equilibrium. It also preserves the ordering of joint policies under the repaired team objective: the shaped team objective $\tilde{J}^{\text{team}}(\pi) = \mathbb{E}_\pi [\sum_{t=0}^{T-1} \gamma^t \sum_{i \in N} \tilde{r}_t^i]$ equals $J^{\text{team}}(\pi) - \rho$ for a nonterminal initial state, since $\sum_i \Phi_0^i = 1$ on a nonempty coalition.*

Proof. For each $i \in N$,

$$\begin{aligned} \sum_{t=0}^{T-1} \gamma^t F_t^i &= \sum_{t=0}^{T-1} (\gamma^{t+1} \Phi_{t+1}^i - \gamma^t \Phi_t^i) \\ &= -\Phi_0^i + \gamma^T \Phi_T^i = -\Phi_0^i. \end{aligned}$$

The identity holds along every trajectory, whatever its length T and whatever the realised scores, because the terminal potential is zero. The initial augmented state is fixed, so the final quantity is independent of every player's policy. Adding a constant to player i 's payoff preserves all unilateral payoff differences and hence its best responses. Applying this to every player preserves Nash equilibria. Summing the componentwise identities gives the team-policy statement. $\square$

This theorem is a proposed repair, not a theorem stated by MARS-RA. Its equilibrium-invariance logic is an instance of the established potential-shaping result for stochastic games [3], and its allowance for potentials that change over time is the dynamic-shaping result of Devlin and Kudenko [1], which MARS-RA cites [4]. What is specific here is the fixed index space and the all-population transition ledger.

4 The active-only boundary

Consider one participation interval of a player i : times $a > 0$ and $b \leq T$ with $i \notin I_{a-1}$, $i \in I_t$ for $a \leq t < b$, and $i \notin I_b$. Under the embedding (1), $\Phi_t^i = 0$

whenever $i \notin I_t$, so the fixed-population ledger pays player i zero on every transition at both ends of which it is inactive. Around the interval it pays two boundary transitions: the entry transition $a-1 \rightarrow a$, which pays $F_{a-1}^i = \gamma\Phi_a^i$, and the exit transition $b-1 \rightarrow b$, which pays $F_{b-1}^i = -\Phi_{b-1}^i$. Over the interval, these payments and the interior ones telescope to

$$\sum_{t=a-1}^{b-1} \gamma^t F_t^i = \gamma^b \Phi_b^i - \gamma^{a-1} \Phi_{a-1}^i = 0. \quad (4)$$

An active-only accounting of the embedded potentials differs from the ledger only in which of the two boundary transitions it pays. Two conventions are natural. If transition t is paid to player i exactly when $i \in I_t$, the exit transition is paid and the entry transition is not, and the recorded shaping sum is

$$\sum_{t=a}^{b-1} \gamma^t F_t^i = \gamma^b \Phi_b^i - \gamma^a \Phi_a^i = -\gamma^a \Phi_a^i. \quad (5)$$

If transition t is paid exactly when $i \in I_t \cap I_{t+1}$, neither boundary transition is paid, and the recorded sum is

$$\sum_{t=a}^{b-2} \gamma^t F_t^i = \gamma^{b-1} \Phi_{b-1}^i - \gamma^a \Phi_a^i. \quad (6)$$

So an entry term $-\gamma^a \Phi_a^i$ appears under both conventions, and an exit term $\gamma^{b-1} \Phi_{b-1}^i$ appears only when the exit transition is unpaid; when it is paid, the exit term vanishes because the potential of an inactive agent is zero. Each surviving term is the negative of a discounted dropped boundary payment, and each can depend on the joint policy. The two kinds of dependence bear on different invariances. The entry time a and the score at entry depend on the actions of the other players before a ; dependence of that kind leaves player i 's unilateral payoff differences unchanged and bears on the ordering of joint policies under the team objective. On a re-entry the entry term also depends on the player's own earlier actions, and the last active state in the exit term depends on the player's own actions; dependence of that kind bears on the player's best responses. The two conventions, and the sorting of the dependences by the invariance they bear on, are a worked illustration of the ledger-level statement that dropped entry and exit payments leave boundary terms that can depend on policy; the case split and the sorting are our elaboration, not a result of the source material. For a first entry the player is inactive at every $t < a$, so under null inactive actions the entry time and entry state can depend only on the other players' policies and the environment. Active-only equivalence therefore requires paying both boundary transitions, which is the fixed-population ledger restricted to its nonzero entries, or compensating transfers that reproduce (4). We do not design such transfers. The exit term in (6) plays, for one player's participation interval, the role of the nonzero terminal potential that Grześ analysed for episodic tasks [2]: the last active state acts as that player's stopping

state, and its potential is not zero. The entry term corresponds to the initial-potential term of that decomposition; there it is fixed by the start state, whereas here the entry time and the entry state can depend on actions.

There is a second interface boundary. MARS-RA begins from a scalar team reward but adds a vector shaping term. One coherent repair declares per-agent payoffs and applies the componentwise theorem. Another uses a scalar team potential, which preserves the ordering of joint policies under the team objective. We examine only one such potential, the coordinate sum of the embedded credits. Since $\sum_i \Phi_t^i = 1$ for every nonempty nonterminal coalition, coordinate-sum scalarisation yields a potential that does not depend on the comparisons, so its shaping signal carries no comparison information. This observation is restricted to coordinate-sum scalarisation. Other scalar functions of the credit vector, such as a weighted sum with unequal weights, are in general not constant, can depend on the comparisons, and still define a scalar potential. We do not analyse them, and we do not exclude comparison-dependent scalar team potentials.

5 Estimation, semantics, and local signal sensitivity

Bradley–Terry scores are defined up to translation, so fix the gauge by taking, for $\hat{c}_t$ and c_t^* , the representatives whose coordinate mean on I_t equals that of v_t^* ; this is the same as minimising the distances below over translations. Define semantic bias

$$\delta_t = \|c_t^* - v_t^*\|_2. \quad (7)$$

Then

$$\|\hat{c}_t - v_t^*\|_2 \leq \|\hat{c}_t - c_t^*\|_2 + \delta_t. \quad (8)$$

Equation (8) is only the triangle inequality, in that common gauge; the bound (12) below holds in any common gauge because softmax is translation invariant. It separates sampling error about the comparator’s latent preference from semantic error to the Shapley target; it does not validate that target or repair the estimator theorem.

For this local comparison, assume that the estimated and target scores at each time are indexed by the same nonempty active coalition I_t . Let

$$d_t = \hat{c}_t - v_t^* \in \mathbb{R}^{|I_t|}, \quad (9)$$

after fixing a common translation gauge. When it appears in the shaping-signal bound below, regard d_t as zero-coordinate embedded in $\mathbb{R}^n$, and set $d_T = 0 \in \mathbb{R}^n$ by convention. Embed the target credits in the same fixed population space by

$$(\Phi_t^*)^i = \mathbf{1}\{i \in I_t\} \text{softmax}_{I_t}(v_t^*)^i, \quad \Phi_T^* = 0, \quad (10)$$

and define the target shaping signal

$$F_t^* = \gamma \Phi_{t+1}^* - \Phi_t^*, \quad (11)$$

writing $\widehat{F}_t = \gamma\Phi_{t+1} - \Phi_t$ for the vector F_t of (3), with a hat only to pair it with F_t^* . The zero-coordinate embeddings preserve Euclidean distances on their common active coalition. Softmax is $1/2$ -Lipschitz in Euclidean norm: its Jacobian at a point with output p is the symmetric positive semidefinite matrix $J = \text{diag}(p) - pp^\top$, and for $\|v\|_2 = 1$ the quadratic form $v^\top J v = \text{Var}_p(v)$ is at most $(\max_i v_i - \min_i v_i)^2/4 \leq 1/2$, so the operator norm of J is at most $1/2$ and the mean value inequality $\|f(x) - f(y)\|_2 \leq \sup\|J\| \|x - y\|_2$ gives the constant. Hence these one-step shaping signals satisfy

$$\|\widehat{F}_t - F_t^*\|_2 \leq \frac{1}{2} (\gamma\|d_{t+1}\|_2 + \|d_t\|_2). \quad (12)$$

This is a local reward-signal bound only. It does not bound the return of a policy produced by MAPPO, the multi-agent proximal policy optimisation learner that MARS-RA uses as its backbone [4], after finitely many updates. Learners can react differently to the same shaped signal, and neither source paper provides the missing stability, regret, or convergence premise.

6 Related work

Lu, Schwartz, and Givigi extend potential-based shaping to multi-player general-sum stochastic games and prove preservation of Nash equilibria [3]. Lu et al. note similar work by Devlin and Kudenko that appeared while their paper was under review; we do not identify it further. Accordingly, the telescoping and general Nash-invariance argument is already published and is not a contribution of this paper.

Two further published results cover steps used here. Devlin and Kudenko, “Dynamic Potential-Based Reward Shaping” (AAMAS 2012, pages 433 to 440), let the potential depend on the time at which a state is visited and show, by the same telescoping along a fixed sequence, that the shaped return differs from the unshaped return only by the initial potential; policy invariance and consistent Nash equilibria therefore survive time-varying potentials [1]. MARS-RA cites this result in support of its own shaping [4]. It covers the clause of our theorem that allows arbitrary realised comparison outputs. Grześ, “Reward Shaping in Episodic Reinforcement Learning” (AAMAS 2017, pages 565 to 573), shows that in episodic tasks the shaped return also contains the discounted potential of the state at which the trajectory stops, that this term depends on actions when termination does, and that unless this potential is zero it can alter the optimal policy and, in a two-agent coordination game, the set of Nash equilibria [2]. The exit term in (6) is a term of this kind for a single player’s participation interval when the exit transition is unpaid, so that part of our boundary diagnosis is a published step applied to one player; the entry term, which appears under both payment conventions, is not covered by it. What is added is the placement of these terms at the entry and exit of individual players under changing coalitions. We read both papers in their author-deposited versions.

Our literature search did not locate a verified publication that states the

particular fixed-population, inactive-coordinate, all-transition ledger construction for changing active coalitions, or that treats individual entry and exit as boundaries of the shaping sum. This is a search result, not a novelty proof.

The two focal papers supply the remaining context. Zhao et al. insist on typed semantic interfaces and explicit downstream composition [5]; Wang et al. supply the changing-coalition comparison and shaping mechanism to which the interface audit applies [4].

7 Interpretation

The repair makes objective-level invariance prediction-independent: once payoff semantics and ledger ownership are fixed, even inaccurate comparisons preserve payoff differences. This should not be read as evidence that comparisons help learning. Under the all-population ledger with a fixed initial augmented state and zero terminal potential, exact shaping changes discounted return only by an initial offset, whether the potential is per-agent or scalar, so any benefit lies on the finite-training optimisation path; under active-only accounting the boundary terms of Section 4 enter the recorded return as well. Under the per-agent repair that path is driven by redistributed per-agent signals. We do not claim that this is the only route: the coordinate-sum observation does not rule out a comparison-dependent scalar team potential, and we have not examined its effect on training. Better estimation can reduce one component of signal error while sharpening a semantically biased target.

8 Limitations and open questions

The result is algebraic and conditional on declared per-agent payoffs, a fixed initial augmented state, a terminal zero potential, null inactive actions, and an all-population transition ledger. It is undefined when $I_t = \emptyset$, or when the score estimate does not exist, as an unregularised maximum-likelihood estimate may not under separation, unless the model declares a fallback; the theorem needs only some declared simplex-valued credit on I_t at every nonterminal step, and an almost surely finite termination time. It does not establish that the published MARS-RA implementation uses this ledger, nor which boundary transitions that implementation pays. It does not give compensating entry or exit transfers for active-only storage, and it does not address how MAPPO, or any learner, consumes shaping rewards paid on boundary transitions to agents that take no actions. For stochastic or history-dependent comparison outputs, an equilibrium theorem may require state augmentation beyond the pathwise identity. The scalarisation observation covers only the coordinate sum of the embedded credits; comparison-dependent scalar team potentials are neither excluded nor analysed.

We do not provide a corrected Bradley–Terry concentration rate, a MAPPO stability theorem, a finite-training return bound, or a query-cost trade-off.

These remain open, as do common units for pricing multimodal-model queries against return or sample efficiency, and whether a comparison-dependent scalar team potential gives a useful training signal. The fixed-space theorem is a correctness layer, not evidence of a positive performance gain and not a general theory of open Markov games.

9 Conclusion

Changing coalitions expose a typing and accounting requirement hidden by the usual potential-shaping notation. A fixed-population all-transition ledger restores componentwise telescoping, by the published argument applied in a fixed index space, and preserves payoff differences for arbitrary comparison outputs. Active-only accounting drops the payments at a player’s entry and exit transitions, and the dropped payments survive as boundary terms that can depend on policy; under the two natural payment conventions these are an entry term, and an exit term of the kind already known for nonzero terminal potentials when the exit transition is unpaid. Scalarising the credits by their coordinate sum removes the comparison signal; other scalarisations are not analysed here. Under exact shaping with the all-population ledger, any benefit of comparison-based shaping lies in finite-training dynamics, for which a learner contract is still missing.

References

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